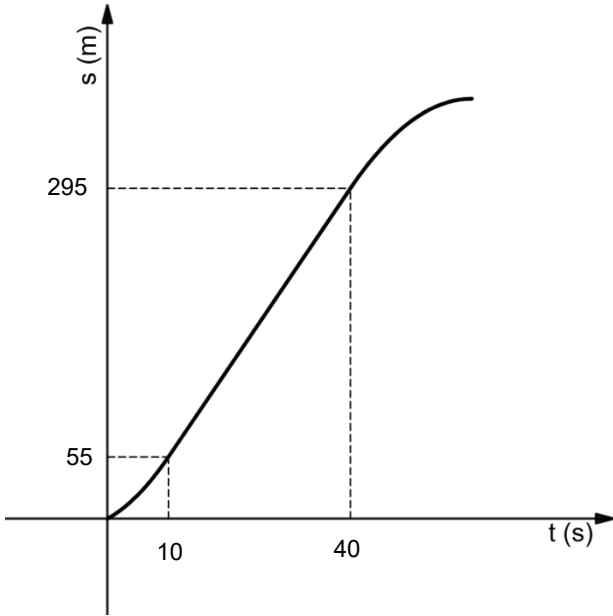




Question	Answer	Marks	Guidance
1	$R = 12g \cos 20$	B1	112.7631145.
	$T - F - 12g \sin 20 = 12 \times 2$	*M1	For use of N2L with 4 terms. Allow sign errors. Allow sin/cos mix. Allow g missing. Allow with their Friction or just F .
	$T - 0.4 \times 12g \cos 20 - 12g \sin 20 = 12 \times 2$ [$T - 45.1052... - 41.0424... = 24$]	DM1	Use of $F = 0.4R$ where R is a component of weight, to get an equation in T only.
	Tension = 110 N [110.147...]	A1	AWRT 110 CWO.
		4	

Question	Answer	Marks	Guidance
2	$P + 5 \sin 60 - 10 \sin 30 = 0$	M1	For resolving in direction of P . 3 terms; allow sign errors and allow sin/cos mix.
	$P = 5 - 5 \frac{\sqrt{3}}{2}$ or 0.670 [0.66987...]	A1	Must be positive.
	$Q = \pm(10 \cos 30 - 5 \cos 60)$	M1	For resolving in direction of Q . 3 terms; allow sign errors and allow sin/cos mix.
	$Q = 10 \frac{\sqrt{3}}{2} - \frac{5}{2}$ or 6.16 [6.16025...]	A1	Must be positive.
		4	

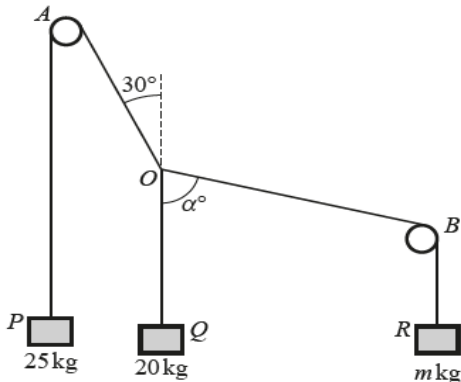
Question	Answer	Marks	Guidance
3(a)	Velocity at $t = 10$ is $8 \text{ [m s}^{-1}\text{]}$	B1	
	$\frac{1}{2} \times 8(T - 40) = 80$ OR $\frac{1}{2} \times 8 \times t = 80 \text{ } [\Rightarrow t = 20]$ and $T = \text{their } t + 40$	M1	Attempt to find time for deceleration. Using <i>their</i> 8.
	Total time = 60s	A1	
		3	
3(b)	Distance after 10s = 55m	B1FT	FT <i>their</i> 8 $15 + 5v$.
	Distance after 40s = 295m	B1FT	FT <i>their</i> 8 $15 + 35v$.
		B1	For at least one quadratic the right way up or a straight line with positive gradient between $t = 10$ and $t = 40$.
		B1	For fully correct, smooth and continuous at $t = 10$ and $t = 40$. Ignore values on s axis. Ignore curve for $t > 60$ if shown on t -axis.
		4	

Question	Answer	Marks	Guidance
4	Use of Newton's second law for P or Q or system	*M1	Allow g missing. Correct number of terms. Allow sin/cos mix. Allow sign errors. Allow using $\theta = 24^\circ$ or better.
	$0.6g \sin 30 - T = 0.6a$ $T - 0.3g \times 0.4 = 0.3a$ $0.6g \sin 30 - 0.3g \times 0.4 = (0.3 + 0.6)a$	A2	A1 For any one correct equation. A2 For any two correct equations.
	For attempt to solve for T or a	DM1	Must get to ' $T =$ ' or ' $a =$ '. From equations with the correct number of relevant terms.
	$a = 2 \text{ ms}^{-2}$ $T = 1.8\text{N}$	A1	Allow ± 2 .
		5	

Question	Answer	Marks	Guidance
5	Attempt at Newton's Second law on either van, trailer or the system	*M1	Must have correct number of terms. Allow sign errors. Dimensionally correct.
	$T - 120 = 350 \times 0.4$ $2500 - X - T = 4500 \times 0.4$ $2500 - X - 120 = (4500 + 350) \times 0.4$	A1	Any 2 correct equations.
	Attempt to solve for either T or X	DM1	From equation(s) with correct number of dimensionally correct terms. Must get $T =$ or $X =$.
	$T = 260 \quad X = 440$	A1	Both correct.
		4	

Question	Answer	Marks	Guidance
6(a)	Velocity at $t = 20$ is $\left[5 + \frac{3}{4} \times 20 = \right] 20 \text{ m s}^{-1}$	B1	Allow if seen on diagram.
	Speed at $t = 50$ is 10 ms^{-1}	B1	When v is minimum. Allow if seen on diagram. Allow -10 .
	Displacement at $t = 40$ is $\frac{1}{2}(5 + \text{their } 20) \times 20 + \frac{1}{2} \times 20 \times \text{their } 20 [= 450]$ OR $\frac{1}{2}((\text{their } 20 - 5) \times 20) + 20 \times 5 + \frac{1}{2} \times 20 \times \text{their } 20 [= 450]$ OR Displacement at $t = 50$ is $\frac{1}{2}(5 + \text{their } 20) \times 20 + \frac{1}{2} \times (\text{their } 20 - 10) \times 30 [= 400]$	*M1	Correct method to find the displacement up to $t = 40$ or $t = 50$. Follow through <i>their</i> 20.
	$\frac{1}{2} \times (T - 40) \times \text{their } 10 = \text{their } 450$ OR $\frac{1}{2} \times (T - 50) \times \text{their } 10 + \frac{1}{2} \times 10 \times \text{their } 10 = \text{their } 450$ OR $\frac{1}{2} \times (T - 50) \times \text{their } 10 = \text{their } 400$	DM1	For an equation in T (or t) involving their displacement at either $t = 40$ or $t = 50$ and using <i>their positive</i> 10 which must have come from $\left \text{their } 20 + \left(\frac{0 - \text{their } 20}{20} \right) \times 30 \right $. Must lead to a value of $T > 0$ unless correctly recovered.
	$T = 130$	A1	Condone $t = 130$ or 130
		5	

Question	Answer	Marks	Guidance
6(b)	$\text{Acceleration} = \frac{\text{their } 10}{\text{their } 130 - 50}$ OR $0 = (\text{their } (-10)) + a \times (\text{their } 130 - 50)$	M1	Correct method to find the acceleration using their 10 (or -10) and their T – dependent on both M marks in part (a) and must lead to a positive value for the acceleration.
	$\text{Acceleration} = \frac{1}{8} \text{ m s}^{-2} \text{ or } 0.125 \text{ m s}^{-2}$	A1	
		2	

Question	Answer	Marks	Guidance
7	For resolving in either direction to form an equation – diagram for reference: 	*M1	Correct number of terms. Allow sin/cos mix. Allow sign errors. Allow g missing. Equation must be in terms of m and α only (so no marks until tensions replaced). For reference: $T_P = 25g$, $T_R = mg$, $T_Q = 20g$, $T_P \cos 30 = T_R \cos \alpha + T_Q$, $T_R \sin \alpha = T_P \sin 30$.
	$25g \cos 30 - 20g - mg \cos \alpha = 0$ [$216.506\dots - 200 - 10m \cos \alpha = 0$]	A1	A0 if g missing.
	$25g \sin 30 - mg \sin \alpha = 0$ [$125 - 10m \sin \alpha = 0$]	A1	A0 if g missing.
	$\alpha = \tan^{-1} \left(\frac{25g \sin 30}{25g \cos 30 - 20g} \right)$ OR $\alpha = \sin^{-1} \left(\frac{25g \sin 30}{(their\ m)g} \right)$	DM1	For attempt to find α . Must get to ' $\alpha =$ ' Must come from equations with the correct number of relevant terms.
	$mg = \sqrt{(25g \sin 30)^2 + (25g \cos 30 - 20g)^2}$ OR $m = \frac{25g \sin 30}{g \sin (their\ \alpha)}$	DM1	For attempt to find m or mg . Must get to ' $m =$ ' or ' $mg =$ ' OR finding the tension in string OR (for reference if correct is $126.085\dots$) and then using $T_R = mg$. Must come from equations with the correct number of relevant terms.
	$\alpha = 82.5$ and $m = 12.6$	A1	AWRT 82.5, 12.6 ($\alpha = 82.4775\dots$ $m = 12.608\dots$) A0 if g missing from original equations.
		6	

Question	Answer	Marks	Guidance
8(a)	$v = 0.6t + 0.6$	B1	Allow un-simplified.
	Velocity at $t = 4$ is 3 m s^{-1}	B1	
		2	

Question	Answer	Marks	Guidance
8(b)(i)	For attempt at integration of $0.3t^{\frac{1}{2}}$ to $kt^{\frac{3}{2}}(+c)$	*M1	Answer must be of the form $kt^{\frac{3}{2}}$ where $k \neq 0$ or 0.3 Use of $v = at$ or $v = k_1(t-4)^{\frac{3}{2}} + \dots$ scores M0 . Do not penalise missing c .
	$their\ 3 = 0.2(4)^{\frac{3}{2}} + c$ leading to $c = \dots$ $\left[\Rightarrow v = 0.2t^{\frac{3}{2}} + 1.4 \right]$	DM1	Use of <i>their</i> 3 from part (a) and $t = 4$ to find c (for reference if correct then $c = 1.4$) OR using correct limits of 4 and T in their equation for v e.g. $\left[0.2t^{\frac{3}{2}} \right]_4^T = 0.2 \left(T^{\frac{3}{2}} - 8 \right)$.
	$14.2 = 0.2T^{\frac{3}{2}} + their\ 1.4$	DM1	For equating to 14.2. Allow in terms of t Dependent on both previous M marks OR equivalent e.g. $14.2 - their\ 3 = 0.2 \left(T^{\frac{3}{2}} - 8 \right)$.
	$T = 16$	A1	Allow 16 or $t = 16$.
		4	

Question	Answer	Marks	Guidance
8(b)(ii)	Distance travelled in first 4 seconds = 7.2m	B1	
	$s = 0.08t^{\frac{5}{2}} + 1.4t$	*B1FT	For integrating their v from part (b)(i) correctly which must be of the form $\alpha t^{\frac{3}{2}} + \beta$ with $\alpha, \beta \neq 0$ – allow un-simplified.
	$\left(0.08(16)^{\frac{5}{2}} + 1.4 \times 16\right) - \left(0.08(4)^{\frac{5}{2}} + 1.4 \times 4\right)$ $[= 81.92 + 22.4 - 2.56 - 5.6 = 104.32 - 8.16 = 96.16]$	DM1	Correct use of limits 4 and <i>their</i> 16 (> 4) must be equivalent to $F(\text{their } 16) - F(4)$.
	Distance = 103.36 m	A1	Condone 103 or better CWO, $\frac{2584}{25}$. Do not ISW if, for example, $103.36 + 7.2 = 110.56$. If constant of integration c found, then must be correct, that is $c = -0.96$. If no integration seen than max B1 for 7.2 and B1 for 103 m or better.
		4	

Question	Answer	Marks	Guidance
9	$R = 80g \times 0.6 [= 480]$	B1	Allow $80g \times \cos 53$ (or better for $\theta = 53.1301\dots$)
	$F = 0.1 \times 80g \times 0.6 [= 48]$	*M1	For use of $F = 0.1R$ with $R = 80g \times 0.6$ or $R = 80g \times 0.8$ or equivalent with $\cos 53$ or $\sin 53$ or better.
	$80g \times 0.8 - F = 80a \Rightarrow a = 7.4$	*M1	For attempt to find an equation for a using N2L with 3 terms; allow sign errors; allow sin/cos mix for the weight component with $\cos 53$ or $\sin 53$ or better. Allow F or their F .
	$v^2 = (0 +)^2 \times (their\ a) \times 5$	DM1	For attempt to find v^2 or v using their positive a .
	Velocity = 8.60 m s^{-1}	A1	Allow $\sqrt{74}$ but A0 for 8.6 if 3sf or better (8.6023...) answer not seen.
	Alternative for Q6(a) for candidates who use an energy method		
	$R = 80g \times 0.6 [= 480]$	B1	Allow $80g \times \cos 53$ (or better for $\theta = 53.1301\dots$).
	$F = 0.1 \times 80g \times 0.6 [= 48]$	*M1	For use of $F = 0.1R$ with $R = 80g \times 0.6$ or $R = 80g \times 0.8$, or equivalent with $\cos 53$ or $\sin 53$ or better.
	[Loss in] PE = $\pm 80g \times 5 \times 0.8 [= \pm 3200]$ OR work done [against] friction = $\pm 0.1 \times 80g \times 0.6 \times 5 [= \pm 240]$	B1	Allow $\cos 53$ or better for the 0.6 in the WD against friction term or $\sin 53$ or better for the 0.8 in the PE term.
	$80g \times 5 \times 0.8 - 0.1 \times 80g \times 0.6 \times 5 = \frac{1}{2} \times 80 \times v^2$ [$3200 - 240 = 40v^2$]	DM1	For attempt at work energy equation. 3 relevant terms; allow sign errors; allow sin/cos mix (using 53 or better) but must be dimensionally correct, terms that need a component should have a component. M0 if the distance in the WD against friction term is not 5.

Question	Answer	Marks	Guidance
9	Velocity = 8.60ms^{-1}	A1	Allow $\sqrt{74}$ but A0 for 8.6 if 3sf or better (8.6023...) answer not seen.
		5	